JOEL THOMAS CYRIL

Candidate for Undergraduate Computer Engineering · Embedded Systems, Web & AI

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01 // PROFESSIONAL SUMMARY

Aspiring Computer Engineering undergraduate with deep practical engineering experience across low-power embedded electronics (ESP32, STM32, Sub-GHz LoRa, custom KiCAD PCBs), full-stack software development (React 19, TypeScript, Node.js, PostgreSQL), and applied AI systems. Demonstrates a verified record of scalable real-world impact—engineering and deploying an AI educational assessment portal to **2,000+ active students**, launching a zero-cost community web platform, captaining award-winning competitive robotics teams (**FLL First Runner-Up**), and maintaining elite academic rigor (**100/100 CBSE Mathematics Centum score**).

02 // EDUCATION & ACADEMIC HONORS

Abu Dhabi Indian School (ADIS) — Abu Dhabi, UAE Graduation: 2026

Senior Secondary (Grade XII CBSE) · Concentration in Computer Science, Mathematics & Physics

- ★ **100/100 Perfect Centum Score** in Grade X CBSE Mathematics Board Examination.
- **95/100** in Grade X CBSE English Language & Literature; consistent top academic percentile across high school cohort.

IIT Madras — Centre for Outreach & Digital Education (CODE) Certified 2026

Course Certification: Introduction to Electronic Systems (8-Week Advanced Program)

Covered circuit theorems, op-amps, semiconductor physics, signal filters, and digital logic micro-architectures.

03 // TECHNICAL CAPABILITIES & PROTOCOLS (ATS KEYWORDS)

Languages:	C, C++, Python, TypeScript, JavaScript, SQL (PostgreSQL, SQLite), HTML5, CSS3, JSON.
Embedded & Hardware:	ESP32, STM32 (STM32WLE5), LoRa (Sub-GHz CC1101), KiCAD (Schematic & PCB Layout), FreeRTOS, Brushless Motors, PID Control, I2C, SPI, UART, RS-485, NFC/RFID (PN532), Infrared TX/RX, 3D CAD.
Full-Stack & Cloud:	React 19, Node.js, FastAPI, PostgreSQL, Supabase, Tailwind CSS, Vite, WebSockets, Socket.io, REST APIs, Vercel, Cloudflare.
Systems, Tools & AI:	Linux (Ubuntu Server), Self-Hosted Home NAS, Git, GitHub, Tesseract OCR, Predictive Telemetry, PlatformIO, Recharts.

04 // FEATURED ENGINEERING PROJECTS

ADIS ReadTrack — AI Literacy & Reading Assessment Platform 2024 – Present

Lead Developer & Systems Architect · React 19, TypeScript, Supabase, PostgreSQL, Tesseract OCR, Tailwind CSS

- Consulted directly with high school English teaching faculty to engineer and deploy a school-wide reading analytics portal currently serving **2,000+ active students** across Abu Dhabi Indian School.
- Integrated camera OCR student ID verification and automated summary evaluations, eliminating paper reading logs and reducing teacher grading workload by **75%**.
- Designed gamified inter-class PISA score leaderboards aligned with ADEK targets (480+ benchmark), with Class 12-B achieving **#1 standing (546 pts)** and school-wide physical education reward privileges.

SwapUAE — Community-Driven Zero-Cash Barter Marketplace 2024 – Present

Co-Founder & Systems Engineer · React 19, TypeScript, Node.js, PostgreSQL, WebSockets, Linux NAS

- Architected full-stack trading application enabling students and UAE community members to exchange books, electronics, and supplies without monetary transactions.
- Grew to **100+ registered members**, **110+ active listings**, and **12+ completed verified trades** within 2 weeks of launch with $0.00 marketing budget.
- Engineered a custom self-hosted Linux NAS server constructed from recycled PC hardware to eliminate cloud database fees, ensuring perpetual zero-cost community hosting.

Reaper One — Multi-Protocol Handheld Hardware & RF Instrument 2024 – 2026

Hardware Engineer & Firmware Developer · ESP32, C++, FreeRTOS, KiCAD, Sub-GHz, NFC, Infrared

- Engineered a standalone multi-protocol RF platform integrating Sub-GHz radio (CC1101), NFC/RFID, IR transceivers, Wi-Fi, and Bluetooth LE into a single custom PCB over 1.5 years of prototyping.
- Designed multi-layer KiCAD schematics and soldered surface-mount components, multiplexing SPI, I2C, and GPIO lines under tight physical constraints with an asynchronous C++ FreeRTOS firmware menu.

Pulso.ai & Manarat Al-Qurm (Environmental & Health AI Telemetry) 2024 – 2026

Award-Winning AI Systems · React, TypeScript, STM32WLE5, Sub-GHz LoRa, FastAPI

- **Pulso.ai:** Award-winning web app addressing UN SDG 3 by synthesizing lifestyle telemetry into predictive wellness risk patterns.
- **Manarat Al-Qurm:** Autonomous coastal mangrove buoy streaming 6-parameter Water Quality Indices over LoRa to Raspberry Pi with 48-hr offline AI forecasting.

05 // LEADERSHIP, COMPETITIONS & ATHLETIC EXCELLENCE

Elected Secretary — ADIS Robotics Club `2023 – Present`

Elected to executive leadership following 3 consecutive years of dedicated membership. Supervise weekly technical workshops, maintain lab equipment, and mentor junior students in EV3/SPIKE coding and autonomous navigation.

Team Captain & Lead Engineer — FIRST LEGO League (UNEARTHED) `2025 – 2026`

Directed competition robot chassis mechanical design, mission strategy, and autonomous sensor dead-reckoning; secured **First Runner-Up in the Robot Game** across top Abu Dhabi schools.

Male Varsity Football Starter — ADEK Cup & CBSE Clusters `2024 – Present`

Selected for Abu Dhabi Indian School starting varsity team in the premier ADEK Cup tournament across two consecutive seasons (current 2025–26 campaign active); represented school at the national CBSE Cluster level.

Honors: Curtin University STEM Top 12 | MUN Delegate | Speedcubing Champion `2022 – 2026`

Top 12 Finalist Team in Curtin STEM Challenge Dubai (2026) · Model United Nations committee delegate (2024–25) · 1st Place Champion in ADIS Rubik's Cube Speedcubing Championship (2022).